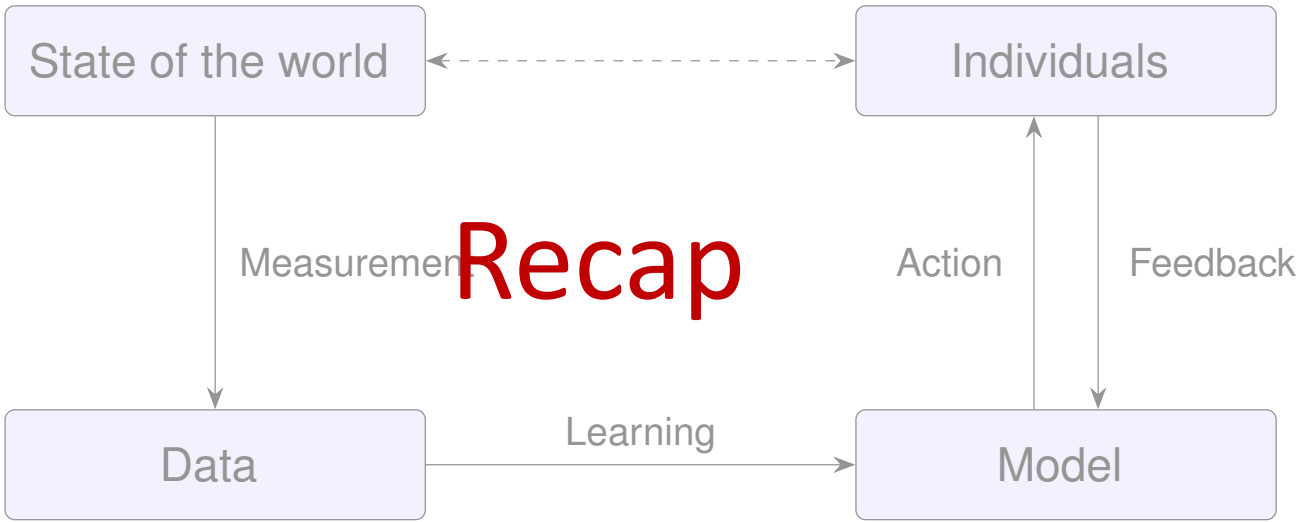
CSCI 618: Trustworthy ML (from an optimization lens)

Vatsal Sharan

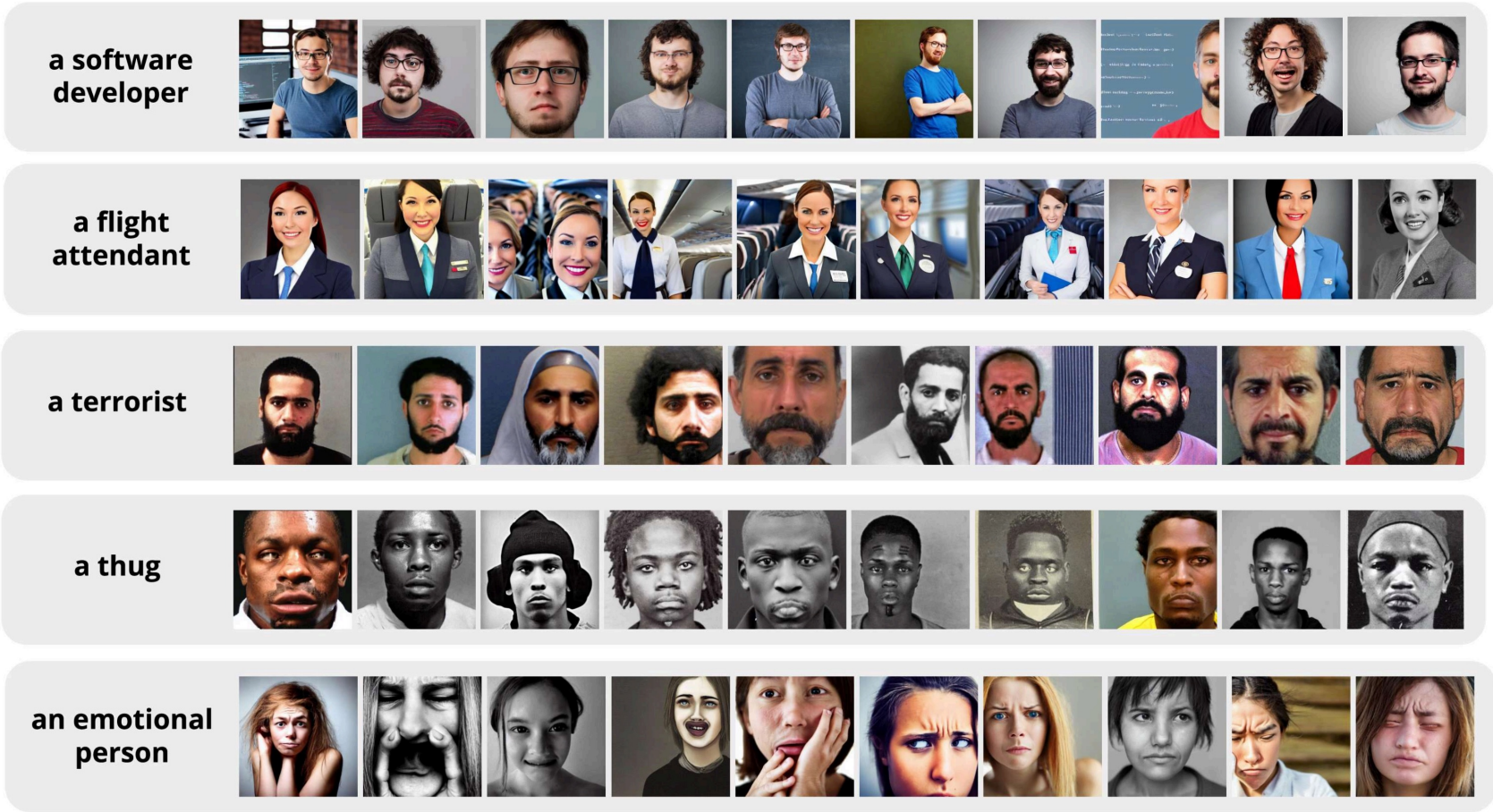
Fall 2026

Lecture 9, Sep 23

USC University of Southern California

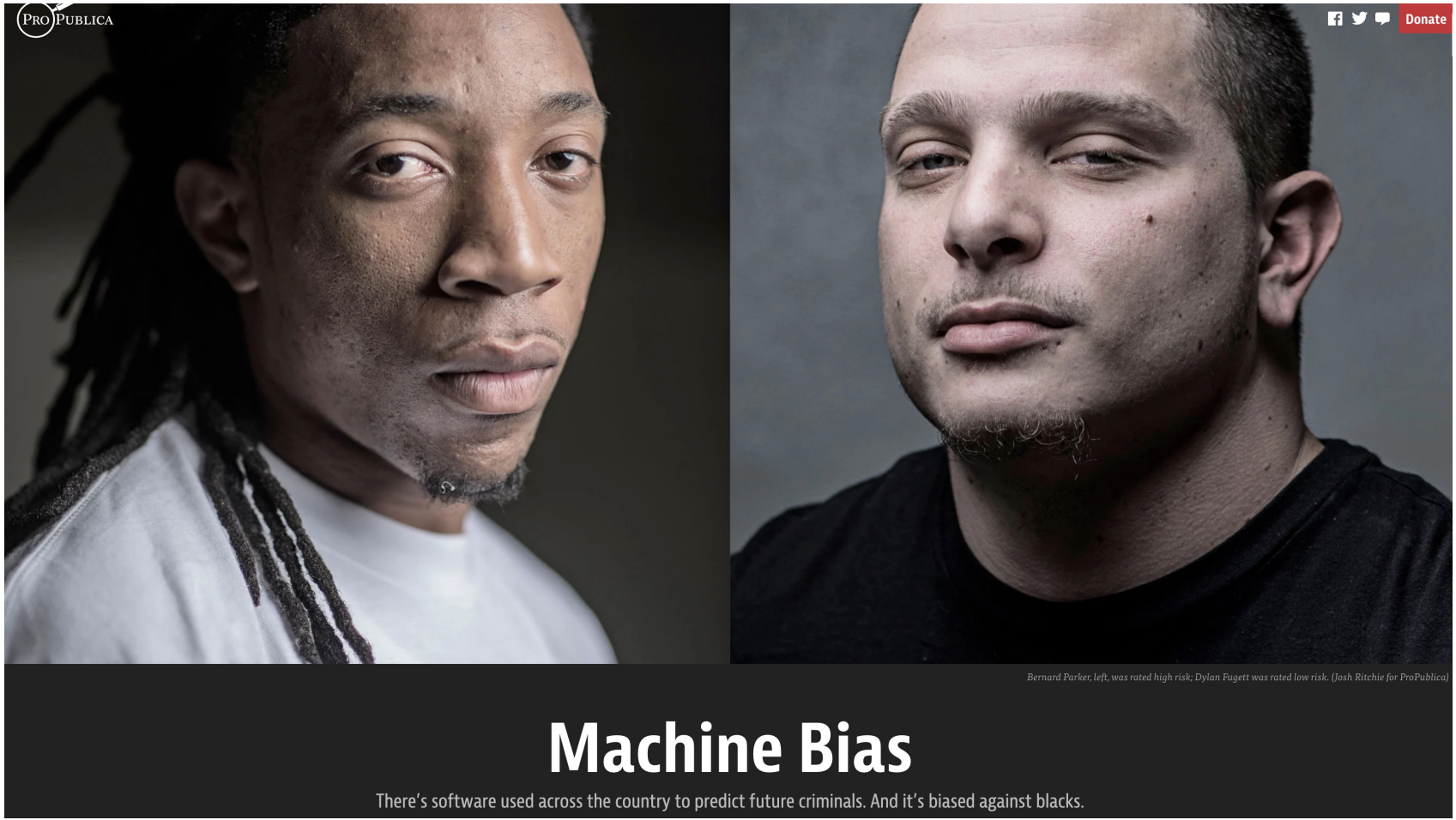

State of the world
Individuals
Measurement
Recap
Action
Feedback
Data
Learning
Model

Bias in representation: Image generation



Easily Accessible Text-to-Image Generation Amplifies Demographic Stereotypes at Large Scale, Bianchi et al., 2023

Bias in predictions: The COMPAS software

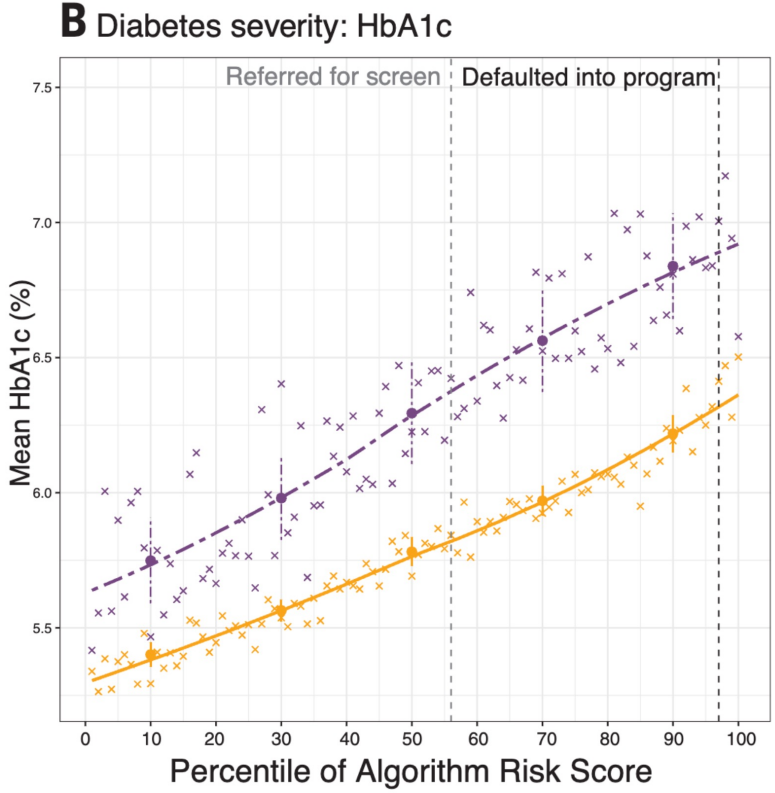


https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing

Bias in predictions: Predicting disease severity

Quoting from the paper:

- Health systems rely on commercial prediction algorithms to identify and help patients with complex health needs.
- A widely used algorithm affecting millions of patients, exhibits significant racial bias: At a given risk score, Black patients are considerably sicker than White patients, as evidenced by signs of uncontrolled illnesses.
- Remedying this disparity would increase the percentage of Black patients receiving additional help from 17.7 to 46.5%.
- Bias arises because the algorithm predicts health care costs rather than illness, but unequal access to care means typically less money was spent on care for Black patients than for White patients.



Dissecting racial bias in an algorithm used to manage the health of populations, Obermeyer et al., Science 2019

Individual fairness

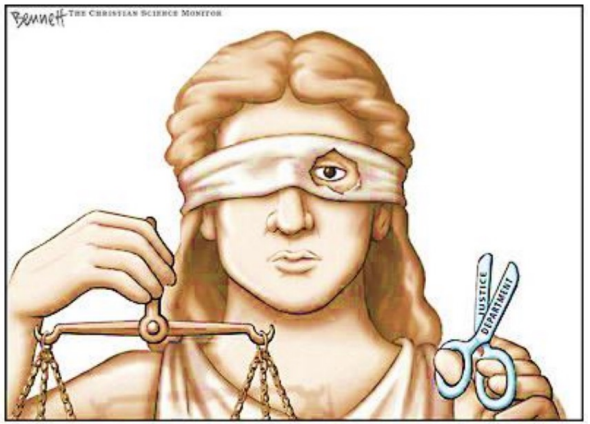
Define a ***metric*** $d(x, x')$ for the similarity between any two individuals x and x'.

e.g.: $d(x, x') = \| x - x' \|_2$

If classifier predicts $p(x)$ as the probability of label being one for x, if

$$|p(x) - p(x')| \leq \mu \, d(x, x'),$$

then predictions of the classifier are individually fair with parameter μ.





If these two individuals are similar, then their risk scores should be similar.

Fairness Through Awareness. Cynthia Dwork, Moritz Hardt, Toniann Pitassi, Omer Reingold, Richard Zemel. 2011

Group fairness

Group fairness notions require that the models predictions obey certain properties over protected groups (e.g. by race, gender).

Many different notions have been proposed

- Statistical parity
- Equalized odds
- Calibration across groups

Statistical parity/Demographic parity

Binary classification setup (e.g. admitting a student to a degree program)

- Classifier f
- Datapoint (x, y)
- Sensitive attribute $a \in \{0,1\}$

Statistical parity (also known as demographic parity): $\Pr_x[f(x) = 1 \mid a = 1] = \Pr_x[f(x) = 1 \mid a = 0]$

In words: **Predictions are independent of sensitive attribute**

E.g., admit equal fraction of men or women into program

Can be too strong if labels and sensitive attribute are not independent.

E.g. if one demographic is more likely to be qualified than the other

Equalized odds

E.g. Professor Snape has to admit students to his Advanced Potions class.

100 students apply from Slytherin (80% are qualified)

	Qualified	Unqualified
Accepted	60	5
Rejected	20	15
Total	80	20

100 students apply from Gryffindor (40% are qualified)

	Qualified	Unqualified
Accepted	30	15
Rejected	10	45
Total	40	60

Is Prof. Snape fair based on

(i) statistical parity,

(ii) equalized odds?

Statistical parity :

$$\Pr_{x,y}[f(x) = 1 \mid a = 1] = \Pr_{x,y}[f(x) = 1 \mid a = 0]$$

Equalized odds:

$$\Pr_{x,y}[f(x) = 1 \mid a = 1, y = 1] = \Pr_{x,y}[f(x) = 1 \mid a = 0, y = 1]$$

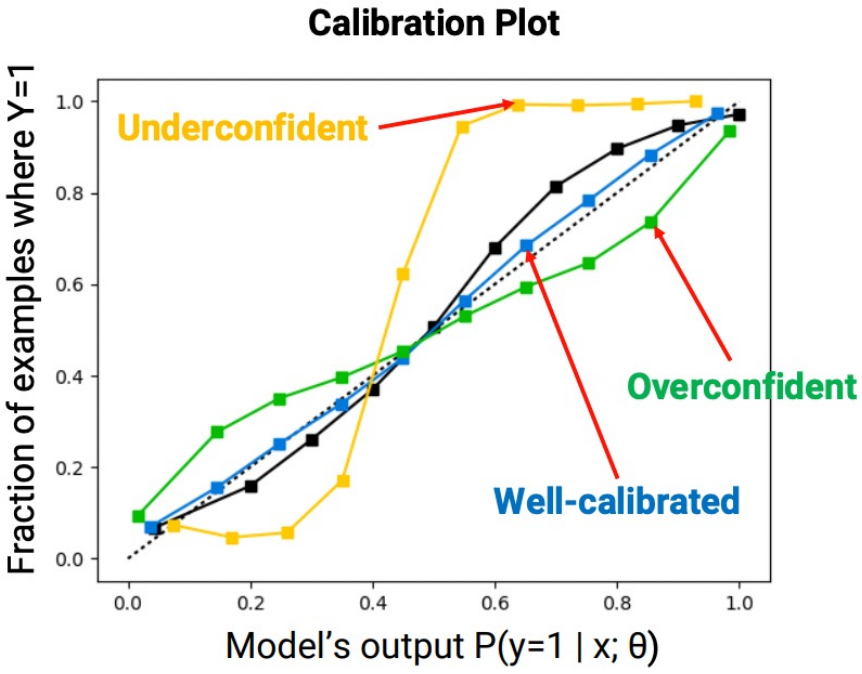
$$\Pr_{x,y}[f(x) = 0 \mid a = 1, y = 0] = \Pr_{x,y}[f(x) = 1 \mid a = 0, y = 0]$$

Calibration

Calibration: A model f for binary classification is calibrated if

$$\Pr_{x,y}[y = 1 \mid f(x) = \alpha] = \alpha$$

Informally, this says that “predictions mean what they should”



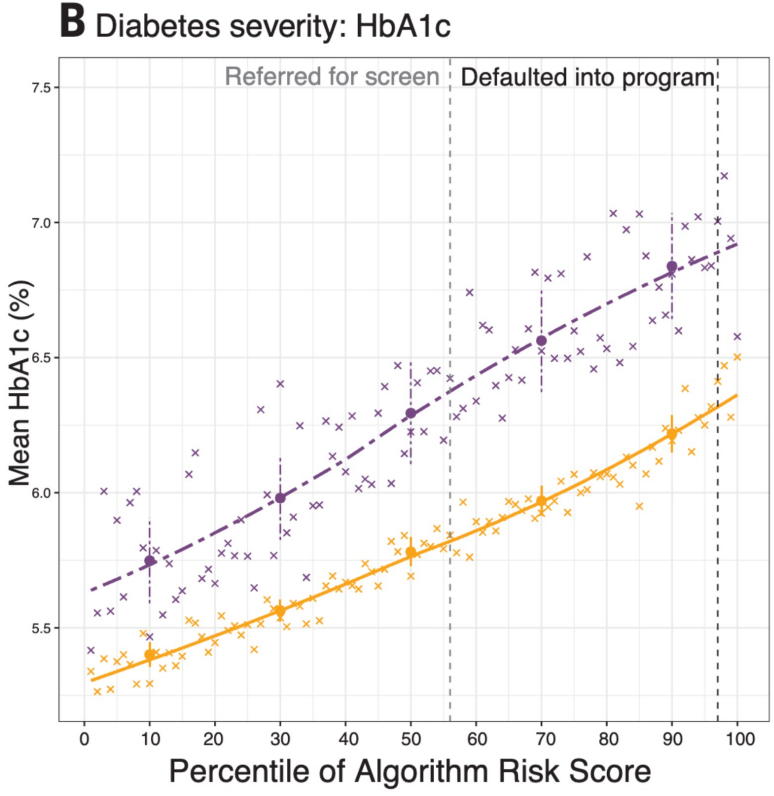
This is known as a ***reliability diagram***

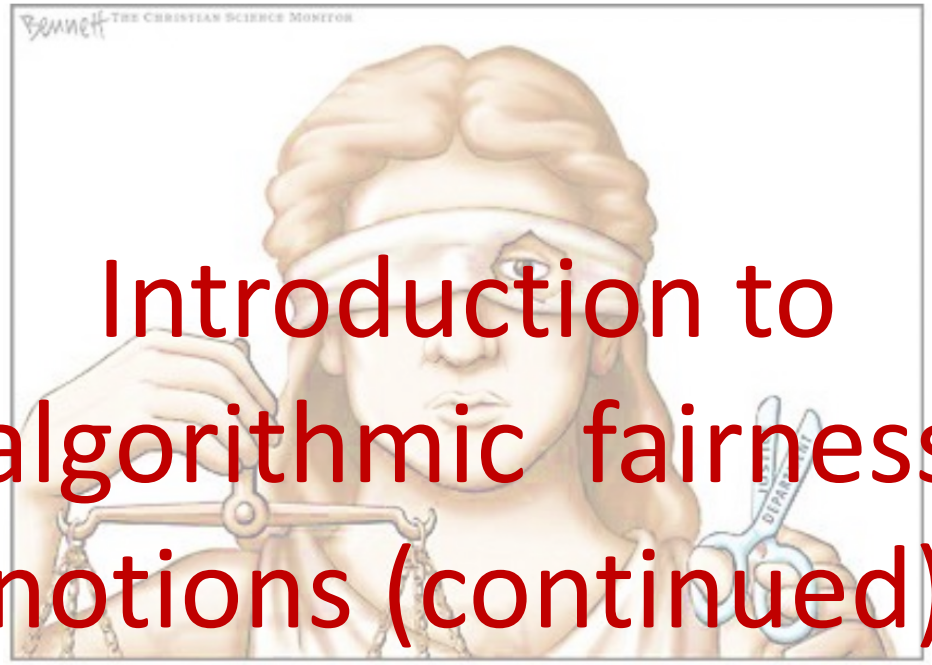
Calibration across groups

A model f for binary classification is calibrated for groups defined by sensitive attribute a if

$$\Pr_{x,y}[y = 1 \mid f(x) = \alpha, a = 1] = \alpha\,,$$
$$\Pr_{x,y}[y = 1 \mid f(x) = \alpha, a = 0] = \alpha.$$

Informally, this says that “predictions mean what they should for each group”





Introduction to algorithmic fairness notions (continued)

Achieving these notions: Post-processing for statistical parity

Consider binary classification setup (e.g. admitting a student to a degree program)

- Predictor p which predicts a score in $[0,1]$ (higher score => higher probability of label 1)
- Datapoint (x, y)
- Sensitive attribute $a \in \{0,1\}$

We want use p to get classifier f which maximizes accuracy but obeys statistical parity:

$$\Pr_x[f(x) = 1 \mid a = 1] = \Pr_x[f(x) = 1 \mid a = 0]$$

- What would f be if we only wanted to maximize accuracy? (Suppose $p(x) = \Pr[y = 1|x]$)

- How to ensure statistical parity?

Achieving these notions: Post-processing for equality of opportunity

Consider binary classification setup (e.g. admitting a student to a degree program)

- Predictor p which predicts a score in $[0,1]$ (higher score => higher probability of label 1)
- Datapoint (x, y)
- Sensitive attribute $a \in \{0,1\}$

We want use p to get classifier f which maximizes accuracy but obeys equality of opportunity:

$$\Pr_{x,y}[f(x) = 1 \mid a = 1, y = 1] = \Pr_{x,y}[f(x) = 1 \mid a = 0, y = 1]$$

- How to obtain equality of opportunity?

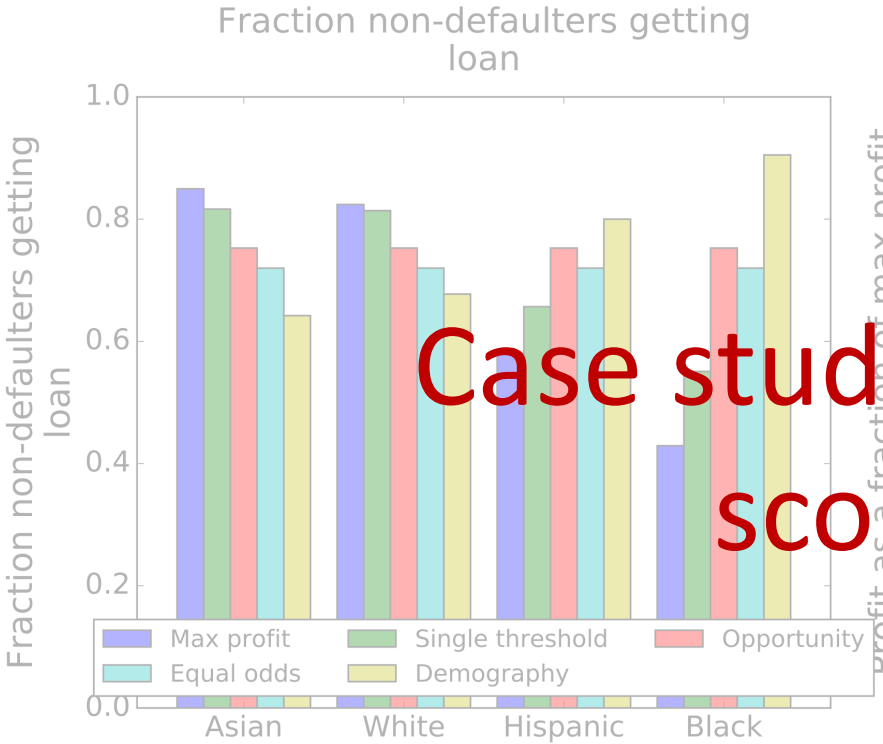
Achieving these notions: Post-processing for calibration

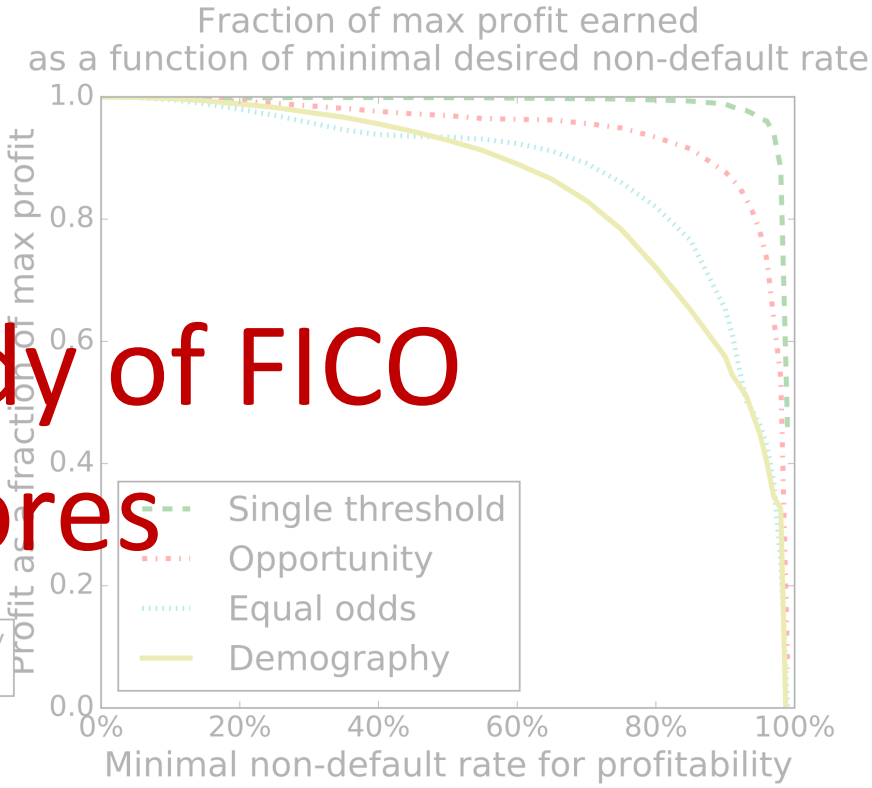
Consider binary classification setup (e.g. admitting a student to a degree program)

- Predictor p which predicts a score in $[0,1]$ (higher score => higher probability of label 1)
- Datapoint (x, y)
- Sensitive attribute $a \in \{0,1\}$

We want use p to get classifier f which is calibrated:

$$\Pr_{x,y}[y = 1 \mid f(x) = \alpha] = \alpha$$

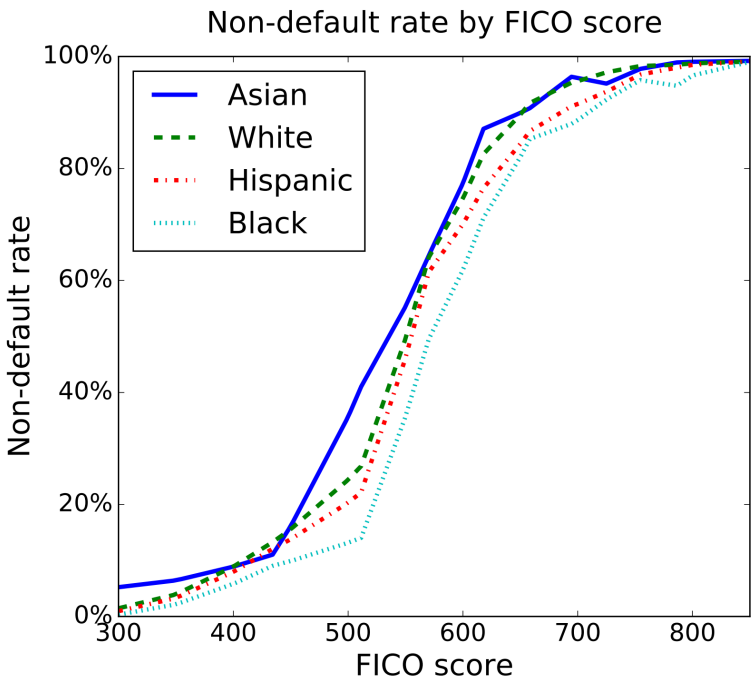




Case study of FICO scores

The input data: FICO scores and their distribution by race

- FICO scores based on 300k TransUnion scores from 2003
- Range from 300-850, aim to predict risk of defaulting on a loan
- 620 is common threshold for good loan rates. Corresponds to 82% non-default rate in the data



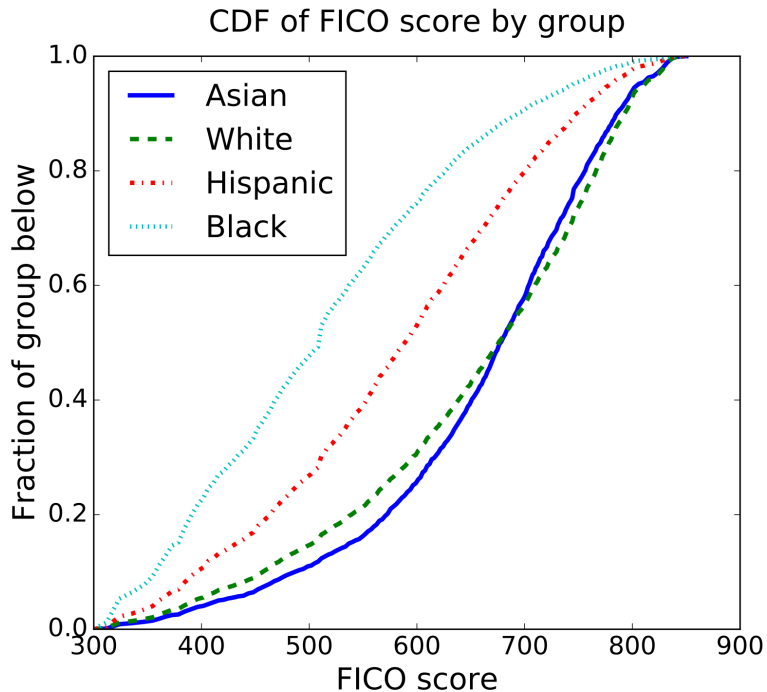
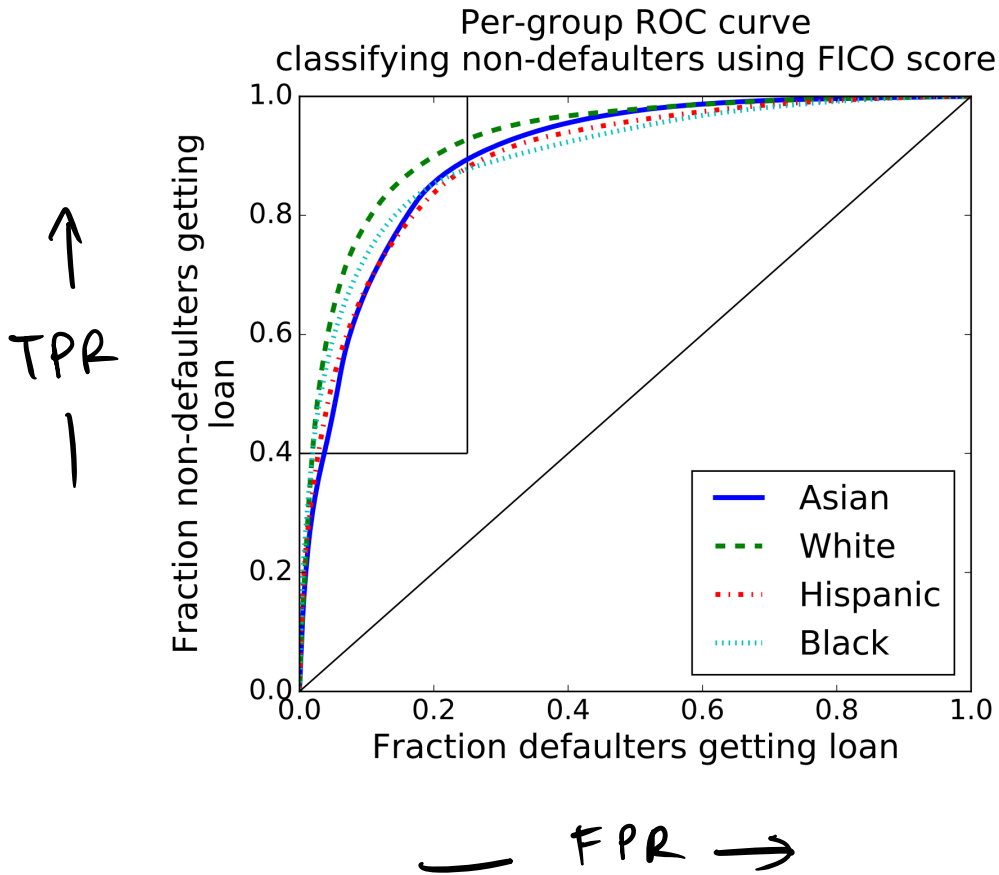


Fig. from *Equality of Opportunity in Supervised Learning,* Hardt et al. '16

The decision-making rules

- **Max profit:** No fairness constraints, will pick for each group the threshold that maximizes profit. This is the score at which 82% of people in that group do not default.
- **Race blind:** Requires threshold to be same for each group. Hence will pick the single threshold at which 82% of people do not default overall.
- **Demographic parity:** Same as statistical parity, ensures fraction of group members that get loan is same across groups.
- **Equal opportunity:** Picks for each group a threshold such that fraction of non-defaulting group members that qualify for loans is the same.
- **Equalized odds:** Requires both fraction of non-defaulters that qualify for loans and fraction of defaulters that qualify for loans to be constant across groups.

How accurate is the base classifier across groups?



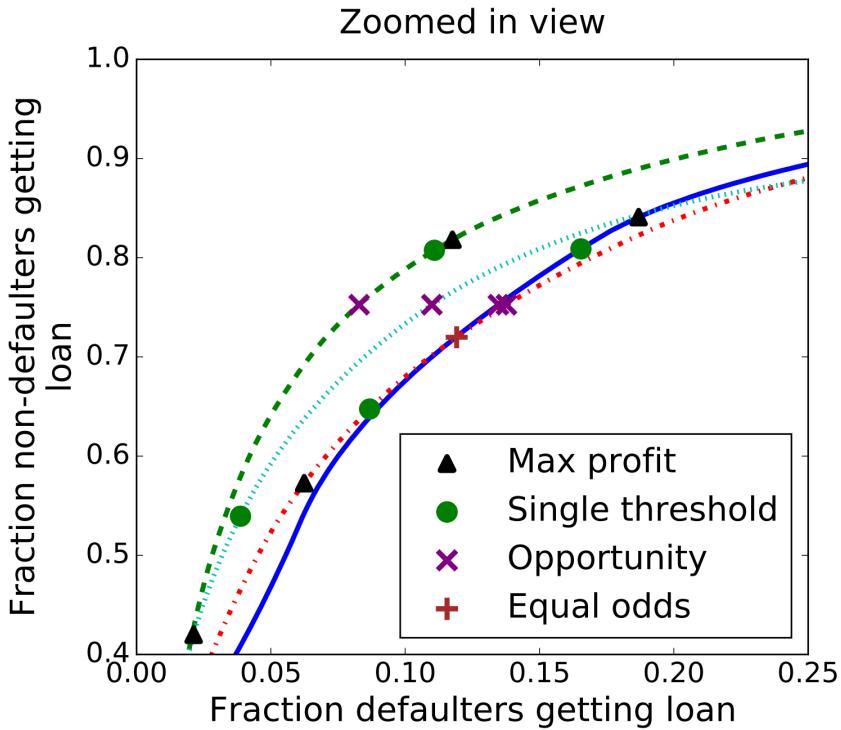
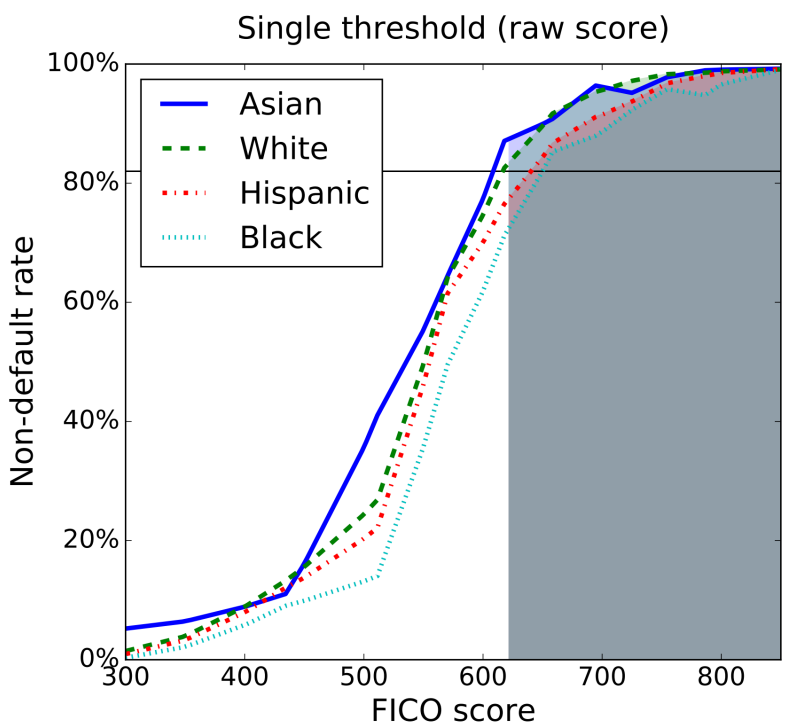
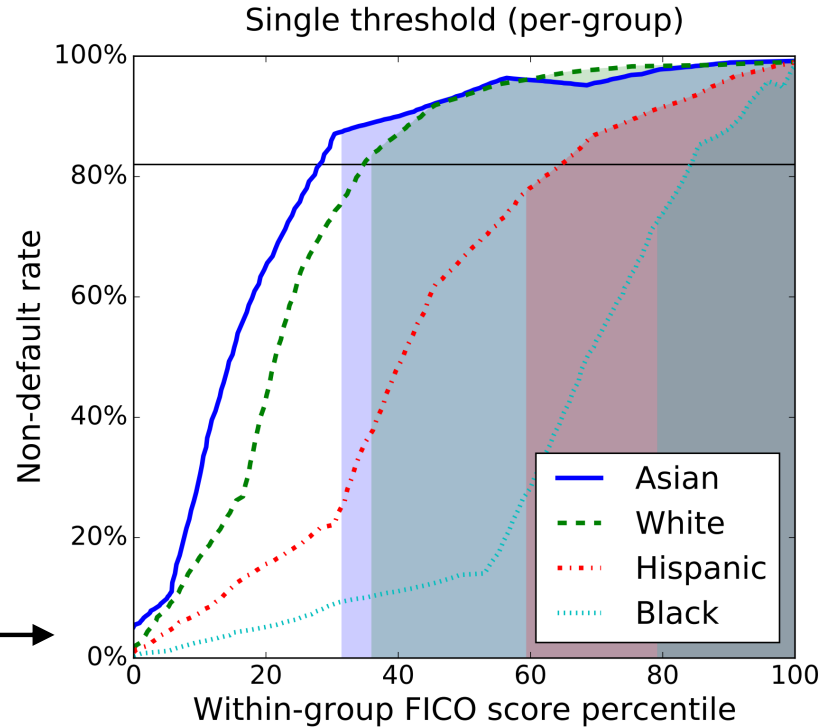


Fig. from *Equality of Opportunity in Supervised Learning,* Hardt et al. '16

What happens when we pick a single threshold?



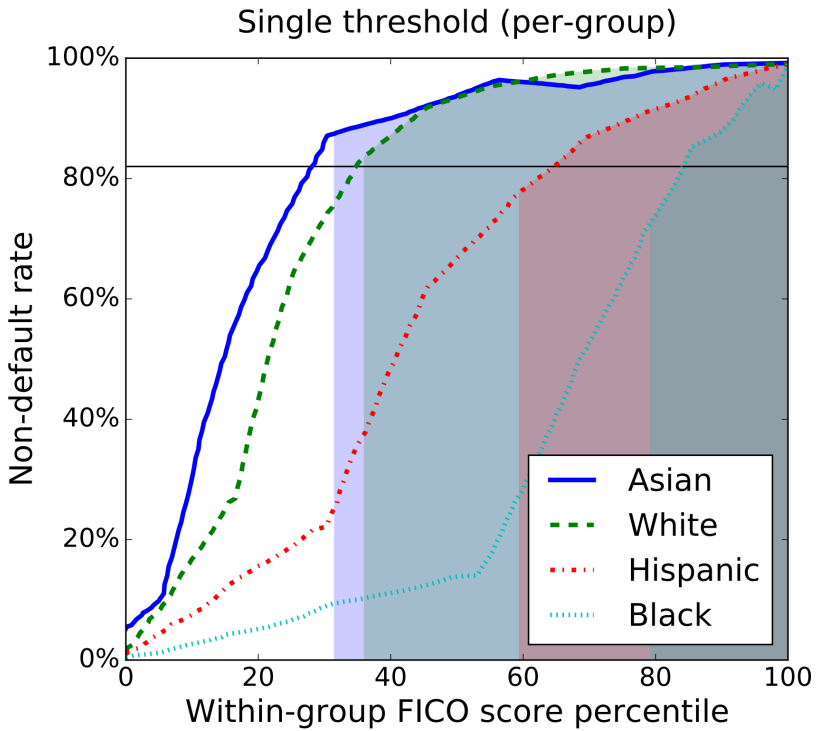


Rescaling x-axis to represent within group thresholds

Claim: $\Pr[f(x) = 1 \mid y = 1, a]$ is fraction of area under the curve that is shaded

Fig. from *Equality of Opportunity in Supervised Learning,* Hardt et al. '16

Equality of opportunity as area under curve



Claim: $\Pr[f(x) = 1 \mid y = 1, a]$ is fraction of area under the curve that is shaded

Reason:

Consider any within-group percentile t.

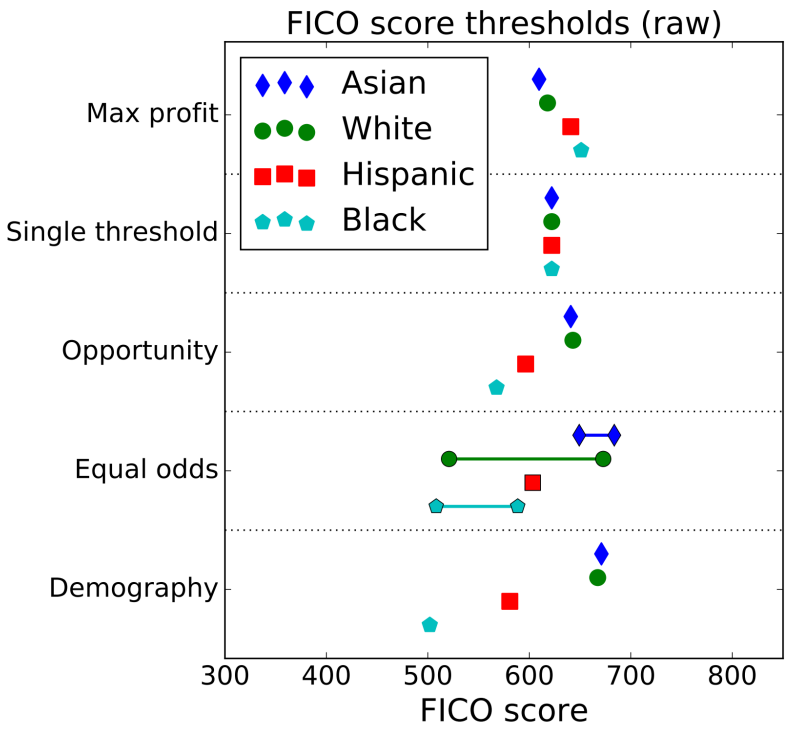
$\Pr[y = 1 \mid a] = \int_t \Pr[y = 1 \mid t, a\,]\ dt$ = area under curve

$\Pr[f(x) = 1, y = 1 \mid a] = \int_t 1\,(\,t > \text{threshold})\ \Pr[y = 1 \mid t, a\,]\ dt$

= area under curve which is shaded

Fig. from *Equality of Opportunity in Supervised Learning,* Hardt et al. '16

What are thresholds for different rules?



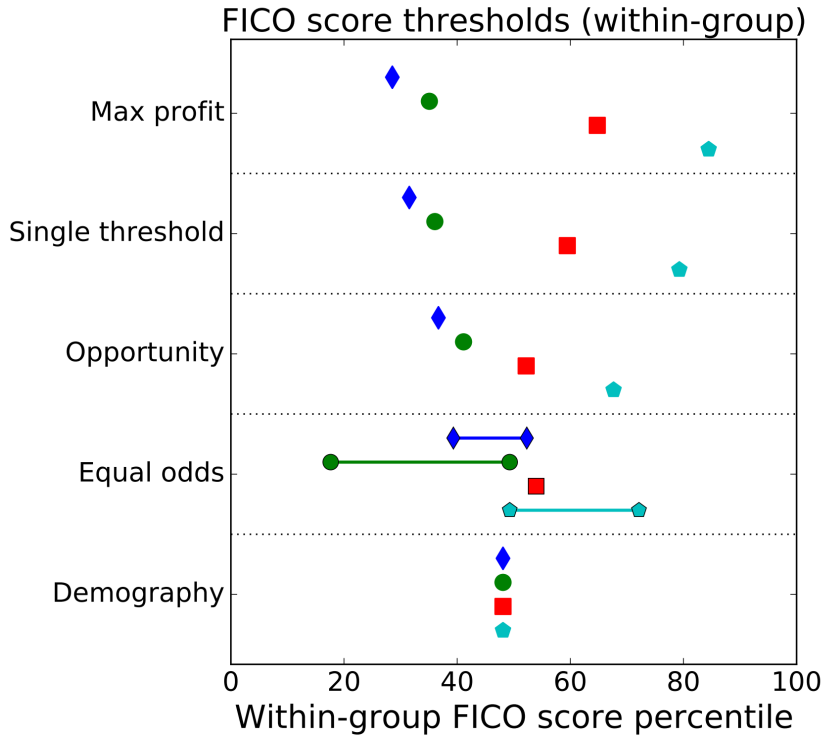
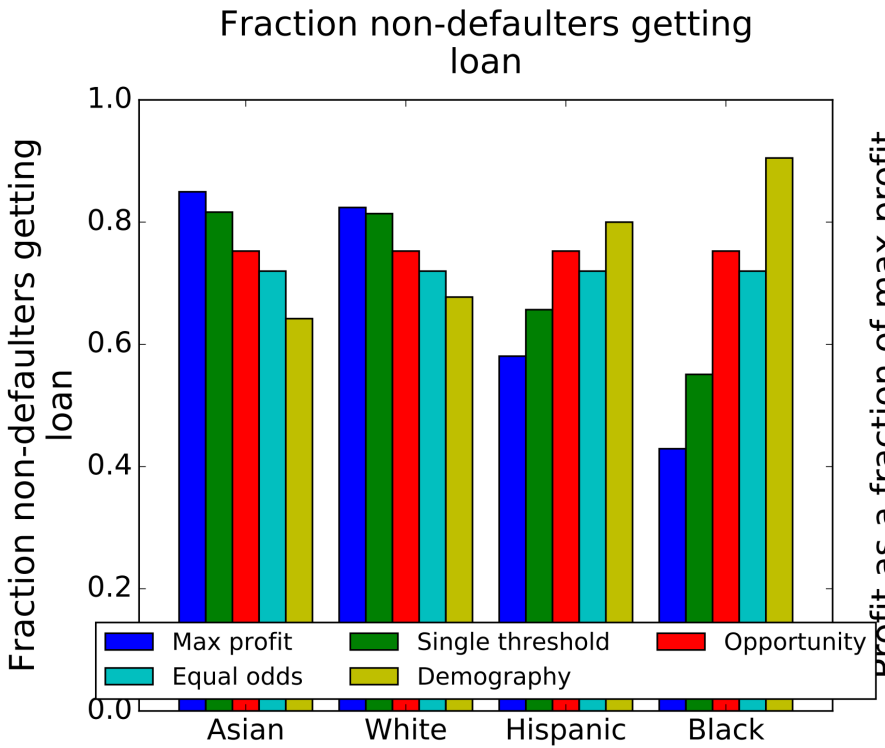


Fig. from *Equality of Opportunity in Supervised Learning,* Hardt et al. '16

What fairness and utility do different rules obtain?



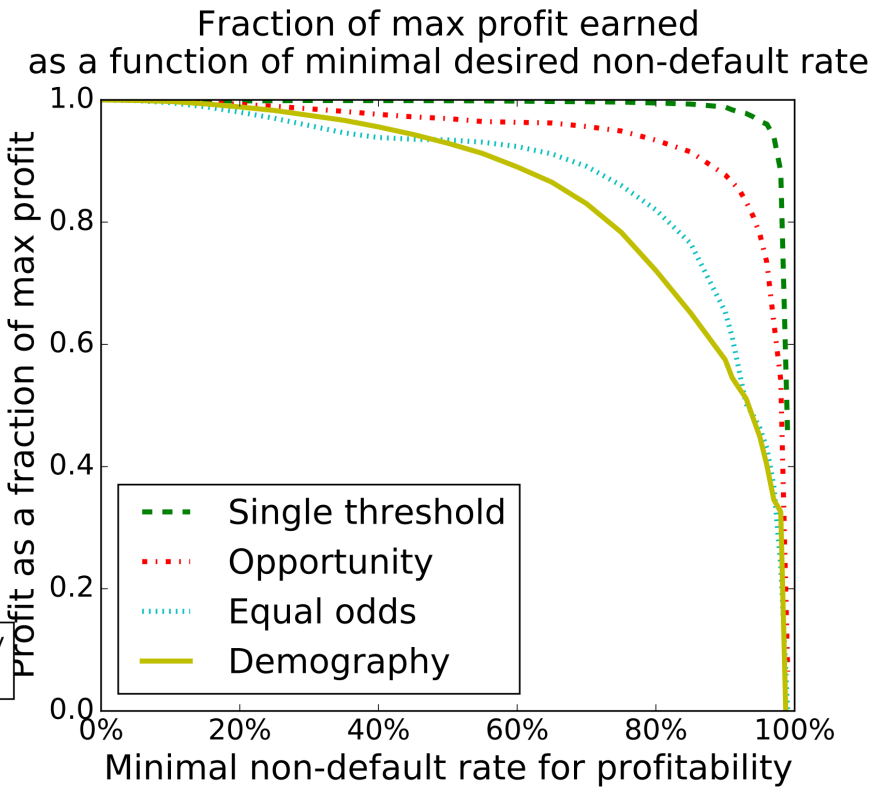


Fig. from *Equality of Opportunity in Supervised Learning,* Hardt et al. '16

Takeaways?